



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-00 Practice Pack

P01-P12 genuine attempts | independent work after teacher walkthrough

Artifact	C3_011_ENGINEERING_FOUNDATIONS_PRACTICE_PACK_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop; book remains sufficient without video.
Bridge boundary	Course 2B competencies may be assumed; all new engineering concepts still start from first principles.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: understand -> follow once -> see expected result -> break/fix -> do a changed version -> validate -> explain -> save evidence. A running command alone never counts as mastery.



Practice rules

The learner book walkthrough comes first. Then close the lesson example as much as practical and do the matching task. Save a genuine attempt before review. Reviews diagnose; they do not replace attempts.

Attempt folder: `evidence/Lxx/attempt\_001/`. Changed retry after review: `evidence/Lxx/retry\_001/`. Never edit an old attempt to hide a mistake.



P01 - Engineering contract

Matches C3-00-L01 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Write a contract for a daily support-ticket summary. Include input/output grain, required fields, business rule, 3 failures, rerun behavior, independent validation, owner and evidence.

REQUIRED EVIDENCE

Save P01 first attempt, three failure scenarios, changed support-ticket contract, one independent validation rule, and a 60-90 second explain-back.

VALIDATE BEFORE REVIEW

- Every requirement can be tested yes/no.
- Input and output grain are explicit.
- At least one independent control would catch a plausible wrong result.
- Failure and rerun behavior are explicit.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P01 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P02 - WSL path lab

Matches C3-00-L02 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Create a Linux-native incoming/processed/archive structure, perform copy/move operations, and save ``pwd``, final tree and command transcript.

REQUIRED EVIDENCE

Save ``wsl -l -v``, ``pwd``, final file tree, ``commands.txt``, changed transfer, and a 60-90 second explanation of Linux-native vs ``/mnt/c`` paths.

VALIDATE BEFORE REVIEW

- Project path is Linux-native.
- Source file you intended to preserve still exists.
- Final tree matches your plan.
- You can explain every relative path from the directory where the command ran.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P02 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P03 - Shell process evidence

Matches C3-00-L03 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

From a changed log, produce level counts and an **ERROR** artifact, preserve source, capture one intentional no-match exit status and reconcile counts.

REQUIRED EVIDENCE

Save shell commands, output files, source/output row controls, one intentional non-zero case, changed transfer, and explain-back.

VALIDATE BEFORE REVIEW

- Source hash is unchanged.
- Counts reconcile to total lines.
- Filtered artifact contains only the requested label.
- Exit status is captured immediately and interpreted correctly.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P03 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P04 - Git reviewable change

Matches C3-00-L04 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Create a feature branch, make two focused commits, prove secret/disposable files are ignored, and write PR summary: what/why/how-tested/risk.

REQUIRED EVIDENCE

Save branch/status/log output, clean secret check, PR-style summary, changed transfer, and explain branch vs commit vs PR.

VALIDATE BEFORE REVIEW

- Branch name and commit history show the intended work.
- No secret/disposable environment is tracked.
- Diff is understood file by file.
- PR summary states what/why/how-tested/risk.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P04 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P05 - Recreatable Python environment

Matches C3-00-L05 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Create venv, install pytest, record dependencies, delete venv, recreate from declarations and prove tests pass.

REQUIRED EVIDENCE

Save version/interpreter output, requirements, first and recreated test output, Git ignore proof, changed transfer and explain-back.

VALIDATE BEFORE REVIEW

- Fresh recreation succeeds.
- Active interpreter lives in project ``.venv``.
- Dependency instructions are sufficient.
- ``.venv`` is ignored by Git.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P05 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P06 - Refactor responsibility boundaries

Matches C3-00-L06 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Refactor the supplied giant script into thin CLI + reusable core + tests. Save before/after structure and import/test proof.

REQUIRED EVIDENCE

Save before/after structure, import smoke check, test output, changed refactor and 60-90 second explanation of module/package/core/CLI.

VALIDATE BEFORE REVIEW

- Core imports without side effects.
- Unit test reaches core without CLI/file-system ceremony where possible.
- CLI delegates rather than duplicates rules.
- Project layout is documented.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P06 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P07 - Configuration precedence

Matches C3-00-L07 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Implement and prove CLI > C3_OUTPUT_DIR > default. Add invalid-config failure and secret boundary evidence.

REQUIRED EVIDENCE

Save three config-resolution runs, ignore/search proof, invalid-config failure, changed transfer and explain-back.

VALIDATE BEFORE REVIEW

- Precedence is documented and demonstrated.
- Invalid values fail before processing.
- No secret is tracked/logged.
- Default is safe and predictable.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P07 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P08 - Operational logging

Matches C3-00-L08 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Produce one successful and one failed run with useful levels/context, non-zero failure, preserved exception evidence and no secret leakage.

REQUIRED EVIDENCE

Save success and failure logs, exit codes, secret scan, changed transfer, and explain why "logged ERROR" is not enough if the process exits 0.

VALIDATE BEFORE REVIEW

- Success/failure exit behavior matches logs.
- Errors include safe actionable context.
- Unexpected exceptions keep trace evidence at boundary.
- No secret/PII dump.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P08 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.

P09 - Testing boundary

Matches C3-00-L09 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Write 4+ tests including edge/failure case; use `tmp_path` for file behavior; inject one deliberate regression and prove tests catch it.

REQUIRED EVIDENCE

Save test file, red-before-green proof for one deliberate regression, final ``pytest -q``, changed transfer, and unit-vs-integration explain-back.

VALIDATE BEFORE REVIEW

- Deliberate regression causes red.
- Clean code causes green.
- Filesystem tests are isolated.
- Test names communicate behavior.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P09 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P10 - Reusable CLI

Matches C3-00-L10 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Build a changed rowcount CLI with --help, stable JSON output, direct core test, and distinct success/failure exit evidence.

REQUIRED EVIDENCE

Save help output, success/failure commands + exit codes, output artifact, core test, changed transfer and explain-back.

VALIDATE BEFORE REVIEW

- Help is understandable.
- Success and failure exit codes differ.
- Output schema/path are stable.
- Core can be tested without CLI parsing.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P10 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.



P11 - Container boundary

Matches C3-00-L11 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

If Docker available: verify engine, build/run a tiny CLI and persist output with a mount. If unavailable: save exact blocker + Dockerfile/commands/expected-result design without claiming runtime success.

REQUIRED EVIDENCE

Save Docker version, real tiny run/build/run evidence OR explicit fallback blocker, Dockerfile, mount explanation, changed transfer and explain-back.

VALIDATE BEFORE REVIEW

- Image vs container explanation correct.
- Client/server engine verified or blocker recorded honestly.
- `--help` works in real container when available.
- Output persistence/mount reasoning is correct.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P11 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.

P12 - Safe rerun + runbook

Matches C3-00-L12 only. Do not open the matching review until this first attempt is saved.

INDEPENDENT TASK

Run unchanged input twice, prove intended result does not multiply, simulate one failure/recovery, and write a complete short runbook.

REQUIRED EVIDENCE

Save run1/run2 comparison, one failure/recovery drill, RUNBOOK.md, changed inventory-snapshot transfer, and explain idempotence vs determinism vs reproducibility.

VALIDATE BEFORE REVIEW

- Rerun rule matches the business/output grain.
- Repeated execution does not multiply intended effect.
- Failure recovery preserves known-good state where contract requires.
- Another operator can follow runbook without hidden steps.

ATTEMPT RECORD

Write: start time, tools used, what you expected, actual result, one uncertainty/mistake, validation performed, and evidence file paths. Status before review must be one of PASS-CANDIDATE / PARTIAL / STUCK - never "mastered".

AFTER YOU SAVE

Open only P12 in C3-012. Record diagnosis, close review, then do the changed retry from the lesson. Mastery requires the changed retry, not recognition of the review answer.